Business Data Analysis and Reporting

➢ Project Overview and Objective

This project focuses on cleaning and organizing raw data in Excel, then using Power BI to build an interactive dashboard. The aim is to show how data preparation and visualization can help in making better business decisions.

➢ Data Sources

- Source: Microsoft Learn (official Power BI documentation)
- Dataset: Financial Sample Excel workbook
- Timeline: Introduced around 2018, continuously available for learning and demo purposes
- Domain: Sales Analytics

➢ Problem Statement

- To analyse sales and profit trends across different years, products, and customer segments.
- To compare revenue performance and discount impact across countries and regions.
- To examine the relationship between units sold, cost of goods sold (COGS), and profitability.
- To understand customer and market segment contributions to overall business growth.

➢ Column Description

Column Name	Description
CustomerID	Unique identifier for each customer.
First_Last Name	Full name of the customer.
Invoice_Number	Unique identifier for each invoice.
Invoice_Date	Date when the invoice was issued.
Product	Name of the product sold.
Segment	Market segment or type of customer (e.g., Government, Corporate, Small Business).
Country	Country where the sale occurred.
Units Sold	Number of product units sold.
Manufacturing Price	Cost to manufacture one unit of the product.
Sale Price	Price at which one unit was sold to the customer.
Discounts	Any discounts applied to the sale.

Sales	Total revenue generated from the sale (Units Sold × Sale Price – Discounts).
COGS	Cost of Goods Sold – total cost of producing the units sold.
Profit	Net earnings from the sale (Sales – COGS).

Cleaned Data Sheet

- Imported the dataset into Excel.
- Added additional columns: **Customer ID, Last_Name, First_Name, Invoice_Number, Invoice_Date**.
- Trimmed and cleaned the columns using Power Query.
- Created a new column **Last_First Name** by merging names with the formula: CONCATENATE([@[last_name]], ", ", [@[first_name]]).
- Corrected the Invoice number format by replacing **INV-11004** with **INV00011004**.
- Standardized the country name by changing **United States of America** to **USA**.
- Adjusted the date format from **01 January 2014** to **01-01-2014** using Power Query.
- Cleaned the **Units Sold** column by removing unnecessary decimals.
- Trimmed and cleaned the numeric columns (**Manufacturing Price, Sale Price, Discounts, Sales, COGS, Profit**) and formatted them with a dollar symbol.

Fact and Dimension Tables

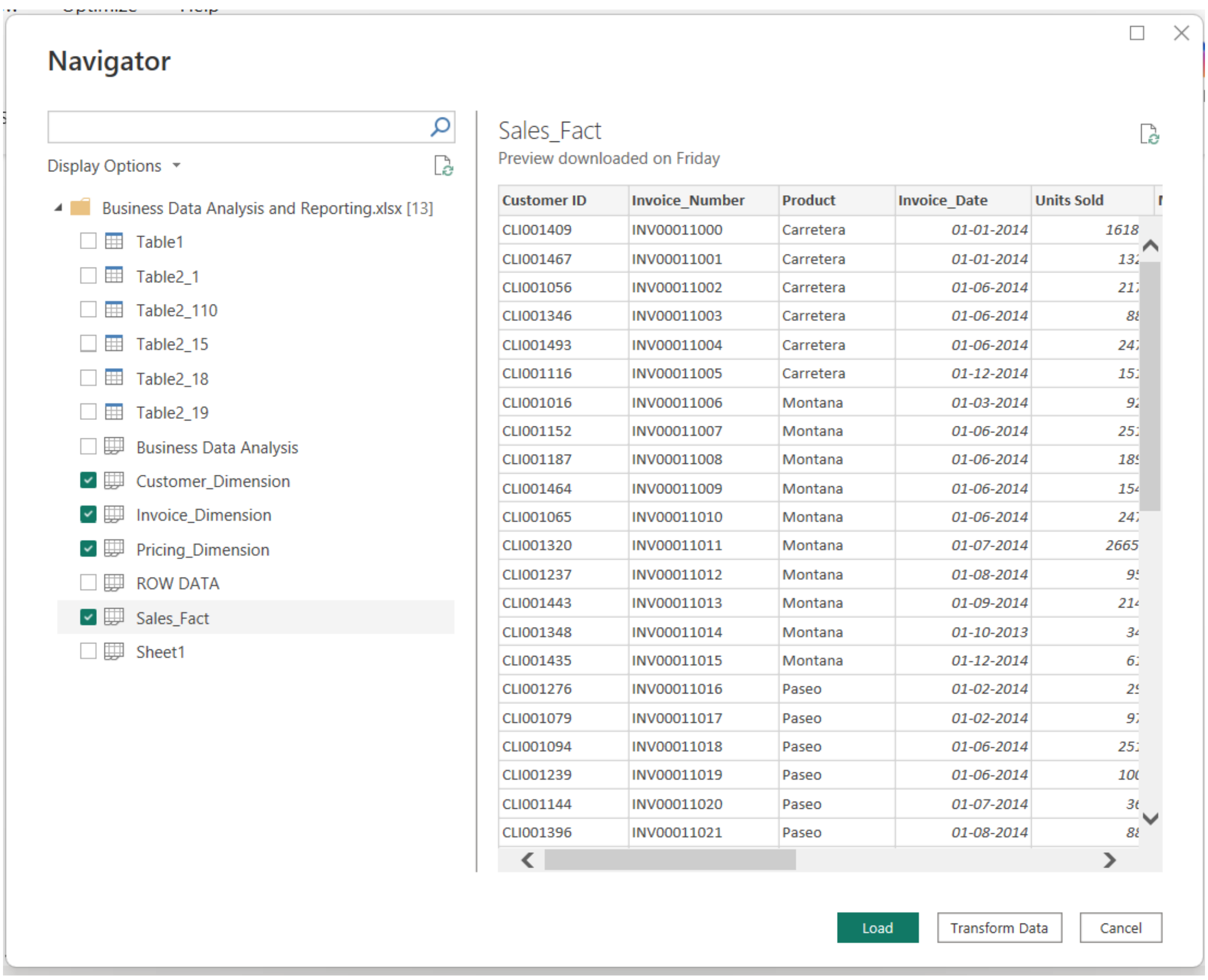
- The columns **Customer ID, Invoice_Number, Invoice_Date, Units Sold, Manufacturing Price, Sale Price, Discounts, Sales, COGS, and Profit** were used to create the **Sales_Fact Table**.
- The columns **Customer ID, First_Last Name, Segment, and Country** were used to create the **Customer Dimension Table**.
- The columns **Product, Units Sold, Manufacturing Price, Sale Price, and Discounts** were used to create the **Pricing Dimension Table**.
- The columns **Invoice_Number, Invoice_Date, and Segment** were used to create the **Invoice Dimension Table**.

Business Data Analysis and Reporting

Power BI Sales Data Model Documentation

➢ Data Loading Process: Excel to Power BI

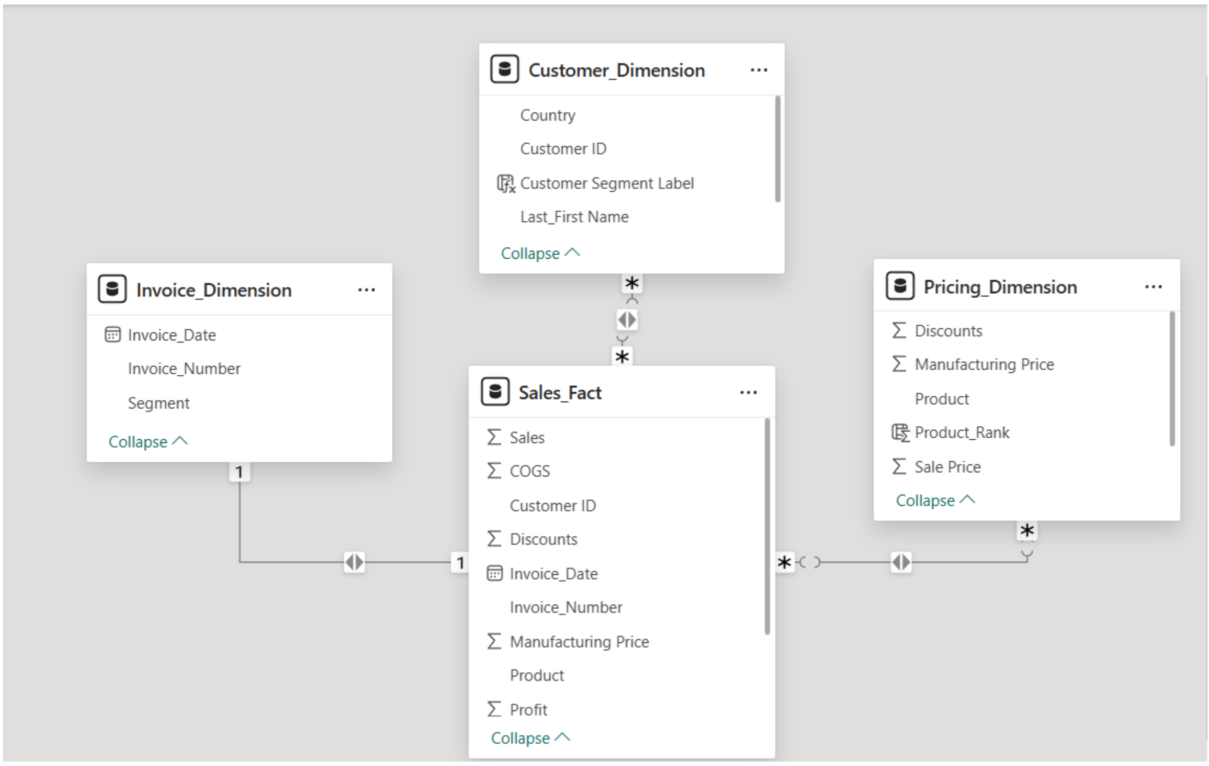
Excel-Driven **Fact** and **Dimension** Tables Imported for Power BI Analysis



Customer ID	Invoice_Number	Product	Invoice_Date	Units Sold
CLI001409	INV00011000	Carretera	01-01-2014	1618
CLI001467	INV00011001	Carretera	01-01-2014	13
CLI001056	INV00011002	Carretera	01-06-2014	21
CLI001346	INV00011003	Carretera	01-06-2014	8
CLI001493	INV00011004	Carretera	01-06-2014	24
CLI001116	INV00011005	Carretera	01-12-2014	15
CLI001016	INV00011006	Montana	01-03-2014	9
CLI001152	INV00011007	Montana	01-06-2014	25
CLI001187	INV00011008	Montana	01-06-2014	18
CLI001464	INV00011009	Montana	01-06-2014	15
CLI001065	INV00011010	Montana	01-06-2014	24
CLI001320	INV00011011	Montana	01-07-2014	2665
CLI001237	INV00011012	Montana	01-08-2014	9
CLI001443	INV00011013	Montana	01-09-2014	21
CLI001348	INV00011014	Montana	01-10-2013	3
CLI001435	INV00011015	Montana	01-12-2014	6
CLI001276	INV00011016	Paseo	01-02-2014	2
CLI001079	INV00011017	Paseo	01-02-2014	9
CLI001094	INV00011018	Paseo	01-06-2014	25
CLI001239	INV00011019	Paseo	01-06-2014	10
CLI001144	INV00011020	Paseo	01-07-2014	3
CLI001396	INV00011021	Paseo	01-08-2014	8

➢ Data Modelling

The data model uses a star schema with the Sales Fact Table at the center holding transaction details. Dimension tables add context: Customer for customer info, Pricing for product and pricing, and Invoice for invoice data.



➢ Calculated Columns

- A calculated column named Customer Segment Label was added to the Customer Dimension Table. It marks customers as Gov Customer if their segment is Government, otherwise as Other Segment.

```
Customer Segment Label = IF(Customer_Dimension[Segment] =
"Government", "Gov Customer", "Other Segment")
```

```
1 Customer Segment Label = IF(Customer_Dimension[Segment] = "Government", "Gov Customer", "Other Segment")
```

Customer ID	Last_First Name	Segment	Country	Customer Segment Label
CLI001409	Guess, Monte	Government	Canada	Gov Customer
CLI001467	Massi Jr, John	Government	Germany	Gov Customer
CLI001116	Kennedy, Sharon	Government	Germany	Gov Customer
CLI001187	Rauscher, Stephen	Government	France	Gov Customer
CLI001443	Robinson, Stephon	Government	Germany	Gov Customer
CLI001276	Smith, Antonio	Government	Canada	Gov Customer

- A calculated column named **Product_Rank** was added to the Pricing Dimension Table. It ranks products by total sales in descending order, using dense ranking.

```
Product_Rank =
RANKX(ALL(Pricing_Dimension),CALCULATE(SUM(Sales_Fact[
Sales])),,DESC,Dense)
```

```
1 Product_Rank = RANKX(ALL(Pricing_Dimension),CALCULATE(SUM(Sales_Fact[ Sales])),,DESC,Dense)
```

Product	Units Sold	Manufacturing Price	Sale Price	Discounts	Product_Rank
Paseo	292	10	20	0	1
Paseo	974	10	15	0	1
Paseo	2518	10	12	0	1
Paseo	1006	10	350	0	1
Paseo	367	10	12	0	1

- A calculated column named Year was added to the Sales Fact Table. It extracts the year from the Invoice Date field

```
Year = YEAR(Sales_Fact[Invoice_Date])
```

```
1 Year = YEAR(Sales_Fact[Invoice_Date])
```

Customer ID	Invoice_Number	Product	Invoice_Date	Units Sold	Manufacturing Price	Sale Price	Discounts	Sales	COGS	Profit	Year
CLI001276	INV00011016	Paseo	01 February 2014	292	10	20	0	5840	2920	2920	2014
CLI001079	INV00011017	Paseo	01 February 2014	974	10	15	0	14610	9740	4870	2014
CLI001094	INV00011018	Paseo	01 June 2014	2518	10	12	0	30216	7554	22662	2014
CLI001239	INV00011019	Paseo	01 June 2014	1006	10	350	0	352100	261560	90540	2014
CLI001144	INV00011020	Paseo	01 July 2014	367	10	12	0	4404	1101	3303	2014

Calculated Tables

- A calculated table named **Country_Product_Sales** was created. It summarizes total sales by combining each customer's country with the product sold.

```
Country_Product_Sales =
SUMMARIZECOLUMNS(Customer_Dimension[Country],Sales_Fact[Product],
"Total Sales",SUM(Sales_Fact[ Sales]))
```

```
1 Country_Product_Sales = SUMMARIZECOLUMNS(Customer_Dimension[Country],Sales_Fact[Product],"Total Sales",SUM(Sales_Fact[ Sales]))
```

Country	Product	Total Sales
Germany	Paseo	13108368.35
Canada	Paseo	14691184.685
Mexico	Paseo	13435246.54
USA	Paseo	14988371.18

- A calculated table named **Top_3_Sales** was created. It returns the three highest sales records from the Sales Fact Table, ordered by sales amount in descending order.

```
Top_3_Sales = TOPN(3,Sales_Fact,Sales_Fact[ Sales],DESC)
```

```
1 Top_3_Sales = TOPN(3,Sales_Fact,Sales_Fact[ Sales],DESC)
```

Customer ID	Invoice_Number	Product	Invoice_Date	Units Sold	Manufacturing Price	Sale Price	Discounts	Sales	COGS	Profit	Year
CLI001145	INV00011423	Velo	01-07-2014 00:00:00	3793.5	120	300	102424.5	1035625.5	948375	87250.5	2014
CLI001476	INV00011405	Montana	01-04-2014 00:00:00	3802.5	5	300	102667.5	1038082.5	950625	87457.5	2014
CLI001424	INV00011192	Paseo	01-07-2014 00:00:00	3450	10	350	48300	1159200	897000	262200	2014

- A calculated table named Total Sales_Table was created. It summarizes each product's total sales and total profit from the Sales Fact Table

```
Total Sales_Table =
SUMMARIZE(Sales_Fact,Sales_Fact[Product],"Total_sales",SUM(Sales_Fac
t[ Sales]),"Total_Profit",SUM(Sales_Fact[Profit]))
```

```
1 Total Sales_Table = SUMMARIZE(Sales_Fact,Sales_Fact[Product],"Total_sales",SUM(Sales_Fact[ Sales]),"Total_Profit",SUM(Sales_Fact[Profit]))
```

Product	Total_sales	Total_Profit
Carretera	13815307.885	1826804.885
Montana	15390801.88	2114754.88
Paseo	33011143.95	4797437.95
Velo	18250059.465	2305992.465
Trail	20511921.02	3034608.02
Amarilla	17747116.06	2814104.06

Measures

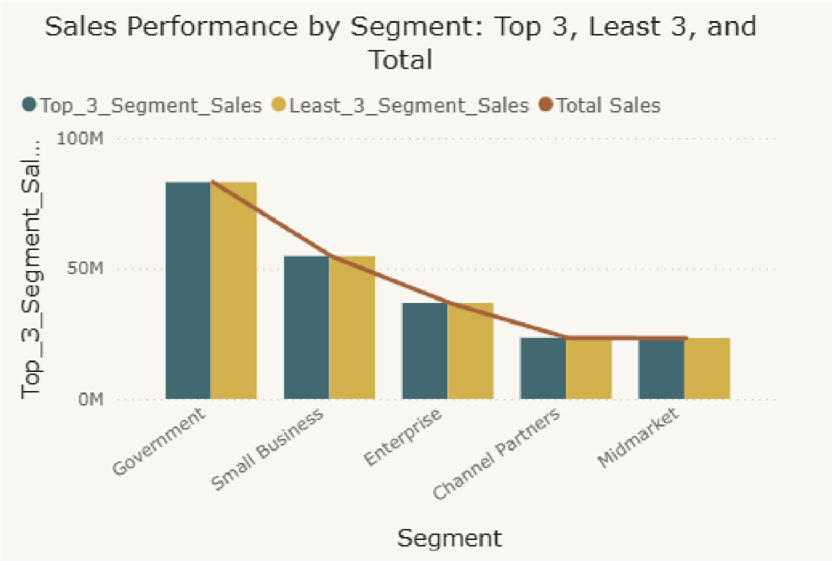
- `Total Sales = SUM(Sales_Fact[ Sales])`
- `Total Profit = SUM(Sales_Fact[Profit])`
-
- `Total COGS = SUM(Sales_Fact[COGS])`
- `Top_3_Segment_Sales = IF(MAX(Customer_Dimension[Segment]) IN SELECTCOLUMNS(TOPN(3, SUMMARIZECOLUMNS(Customer_Dimension[Segment], "Total Sales", [Total Sales]), [Total Sales], DESC), "Segment", Customer_Dimension[Segment]), [Total Sales], BLANK())`
- `Top_3_Product_Sales = IF(MAX(Pricing_Dimension[Product]) IN SELECTCOLUMNS(TOPN(3, SUMMARIZECOLUMNS(Pricing_Dimension[Product], "Total Sales", [Total Sales]), [Total Sales], DESC), "Product", Pricing_Dimension[Product]), [Total Sales], BLANK())`
- `Least_3_Segment_Sales = VAR BottomSegments = TOPN( 3, SUMMARIZECOLUMNS( Customer_Dimension[Segment], "Total Sales", [Total Sales] ), [Total Sales], ASC ) RETURN IF( MAX(Customer_Dimension[Segment]) IN SELECTCOLUMNS(BottomSegments, "Segment",`

```
Customer_Dimension[Segment]), [Total Sales], BLANK() )
```

- ```
  Least_3_Product_Sales = VAR BottomProducts = TOPN( 3,
  SUMMARIZECOLUMNS( Pricing_Dimension[Product], "Total Sales",
  [Total Sales] ), [Total Sales], ASC ) RETURN IF(
  MAX(Pricing_Dimension[Product]) IN
  SELECTCOLUMNS(BottomProducts, "Product",
  Pricing_Dimension[Product]), [Total Sales], BLANK() )
  ```
- ```
  Average sales = AVERAGE(Sales_Fact[ Sales])
  ```
- ```
  Average Profit = AVERAGE(Sales_Fact[Profit])
  ```

Data Visualization

Sales Performance by Segment: Top 3, Least 3, and Total

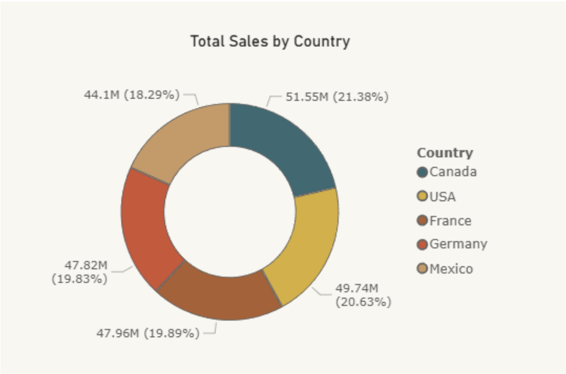


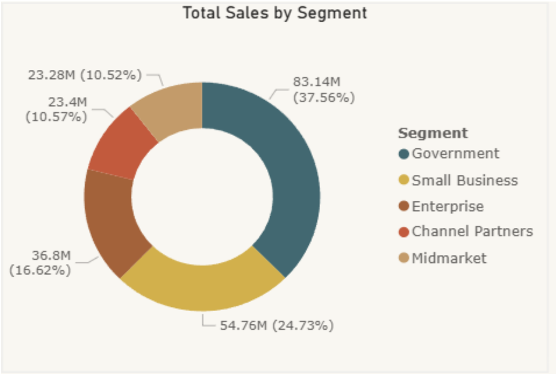
Key Insights

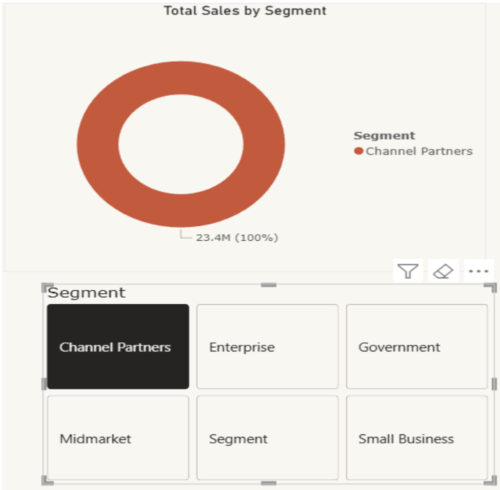
- Government, Small Business, and Enterprise lead sales, while Channel Partners and Midmarket underperform.
- Total sales across all segments reach ₹11,87,26,350.26, with Government and Small Business driving the majority.
- Enterprise shows moderate performance, positioned between high and low contributors.
  ```

Business Data Analysis and Reporting

Sales Analysis by Segment and Country



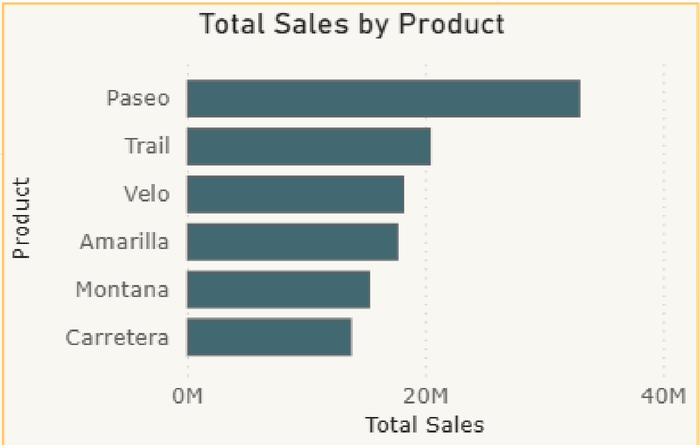




Key Insights

- Government and Small Business segments generate the highest sales.
- Channel Partners, Midmarket, and Mexico show the lowest performance.
- USA and Canada lead in sales contributions, while Mexico lags behind.
- Strong performance is likely driven by larger contracts and wider customer reach.

Product Category Sales and Profit Analysis



Product	Sum of Sale Price
Amarilla	12096
Carretera	10395
Montana	10890
Paseo	21852
Trail	15106
Velo	12561
Total	**82900**

Product	Average of Profit
Amarilla	29,937.28
Carretera	19,643.06
Montana	22,739.30
Paseo	23,749.69
Trail	27,840.44
Velo	21,155.89
Total	**24,133.86**

Key Insights

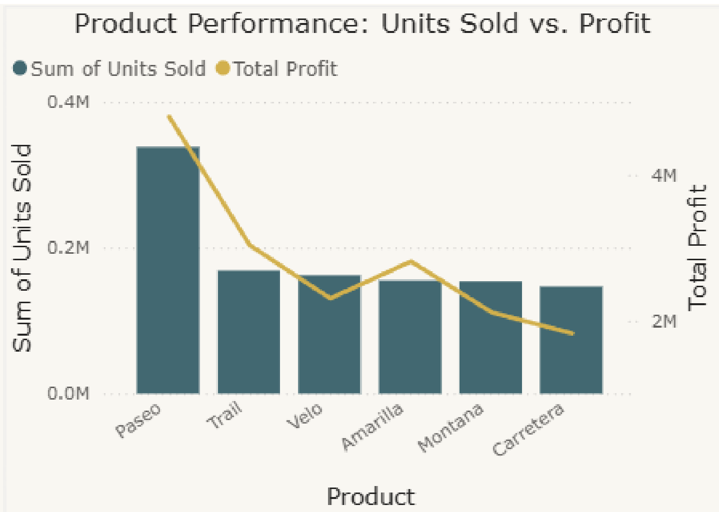
- Paseo leads with the highest sales ($3.30 crore) and profit ($47.97 lakh).
- Carretera shows the lowest profit ($18.26 lakh), despite reasonable sales.
- Overall sales across all categories total $11.87 crore, with profits at $1.68 crore.
- Strong performance in Paseo suggests effective demand and pricing, while Carretera's weak margins point to higher costs.

Total Sales and Profit by Year

Key Insights

- Sales grew from $26.4M in 2013 to $92.3M in 2014, with profits rising from $3.9M to $13M.
- The sharp increase in 2014 indicates stronger demand and expanded market reach.
- Profit growth aligned with sales growth, showing improved efficiency compared to 2013.

Product Performance: Units Sold vs. Profit



Business Data Analysis and Reporting

Key Insights

- Paseo achieved the highest unit sales (338,240) and profit ($4.79M).
- Carretera recorded the lowest profit ($1.82M) despite selling 146,846 units.
- Paseo’s strong sales volume directly drove its high profitability.
- Carretera’s weaker margins or higher costs limited its profit contribution.

Customer-Level Detail

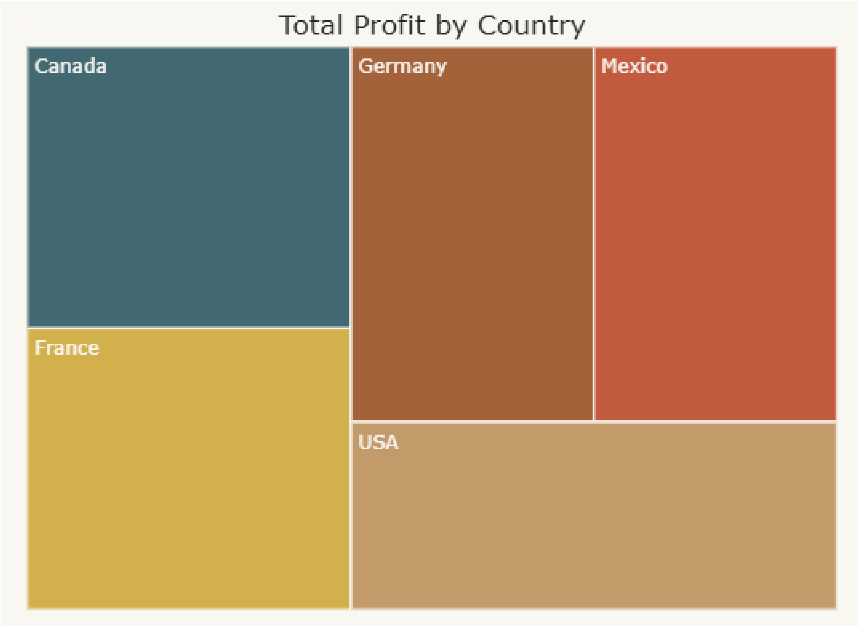
This table shows **customer-level financial details**, where each row lists a customer’s ID along with their total sales, cost of goods sold (COGS), and resulting profit. In short, it highlights how much each customer contributed to revenue and profitability.

Customer ID	Sum of Sales	Sum of COGS	Sum of Profit
CLI001000	1,84,989.00	1,49,370.00	35,619.00
CLI001001	4,89,102.75	4,17,630.00	71,472.75
CLI001002	14,387.40	6,990.00	7,397.40
CLI001006	7,91,628.56	6,45,345.00	1,46,283.56
CLI001007	4,69,916.28	3,82,477.00	87,439.28
CLI001008	8,40,937.50	7,81,340.00	59,597.50
CLI001010	3,29,366.25	2,74,440.00	54,926.25
CLI001011	3,21,429.99	2,70,745.00	50,684.99
CLI001012	9,662.40	6,710.00	2,952.40
CLI001013	57,968.35	37,605.00	20,363.35
CLI001014	6,28,194.75	5,83,500.00	44,694.75
CLI001015	25,134.40	13,660.00	11,474.40
CLI001016	5,23,506.00	4,75,960.00	47,546.00
CLI001017	3,09,028.09	2,62,940.00	46,088.09
CLI001019	89,966.25	87,240.00	2,726.25
CLI001020	15,940.98	4,108.50	11,832.48
CLI001022	44,385.52	13,487.00	30,898.52
CLI001023	58,710.32	28,811.00	29,899.32
CLI001026	1,91,884.00	1,61,980.00	29,904.00
CLI001027	10,98,698.40	9,47,220.00	1,51,478.40

Business Data Analysis and Reporting

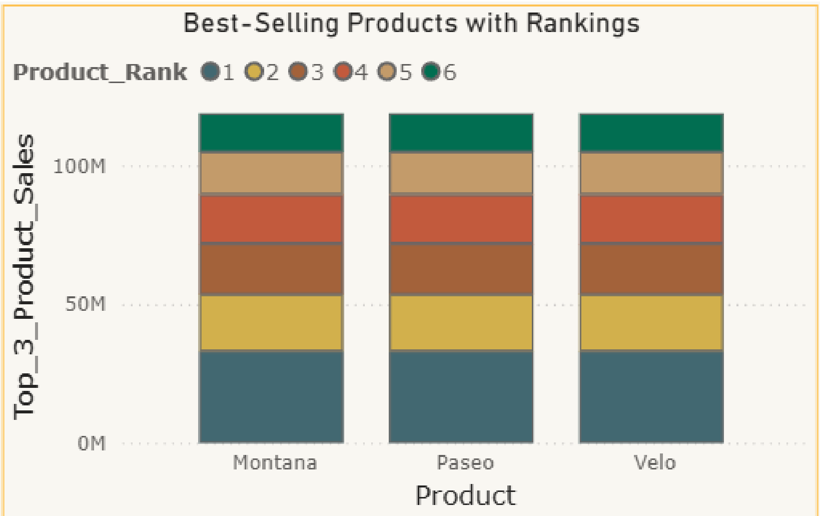
Total Profit by Product (Across Countries)

The treemap highlights that France, Germany, and Canada contribute the highest profits, while Mexico and the USA show relatively lower figures, all together summing up to a grand total of about $168.9 million.



Best-Selling Products with Rankings

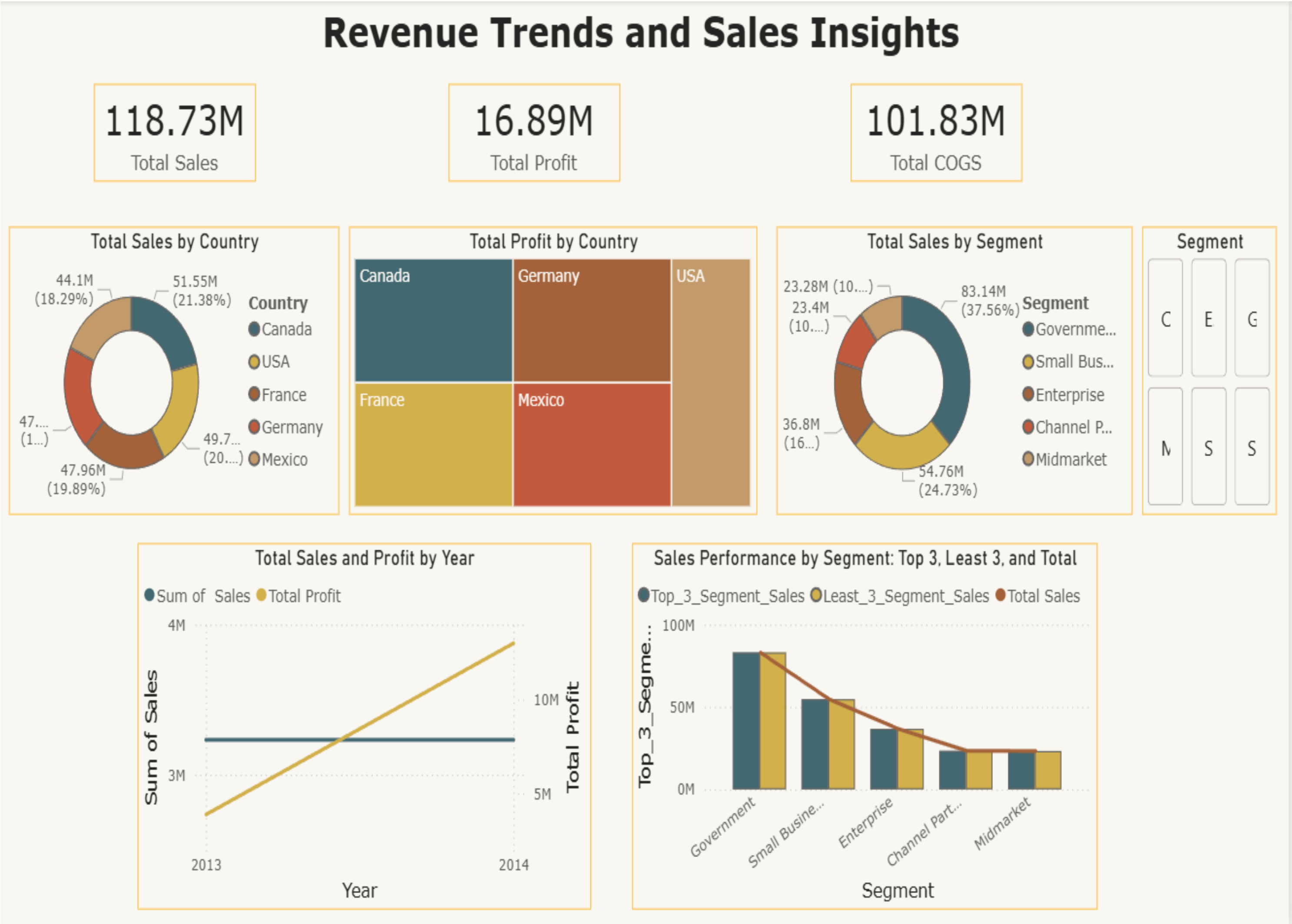
All three products (Montana, Paseo, and Velo) show nearly equal total sales, with Rank 1 and Rank 2 contributing the most significant share. Lower ranks (3–6) add smaller, consistent portions, indicating a balanced and stable sales distribution across all products.



Future Insights

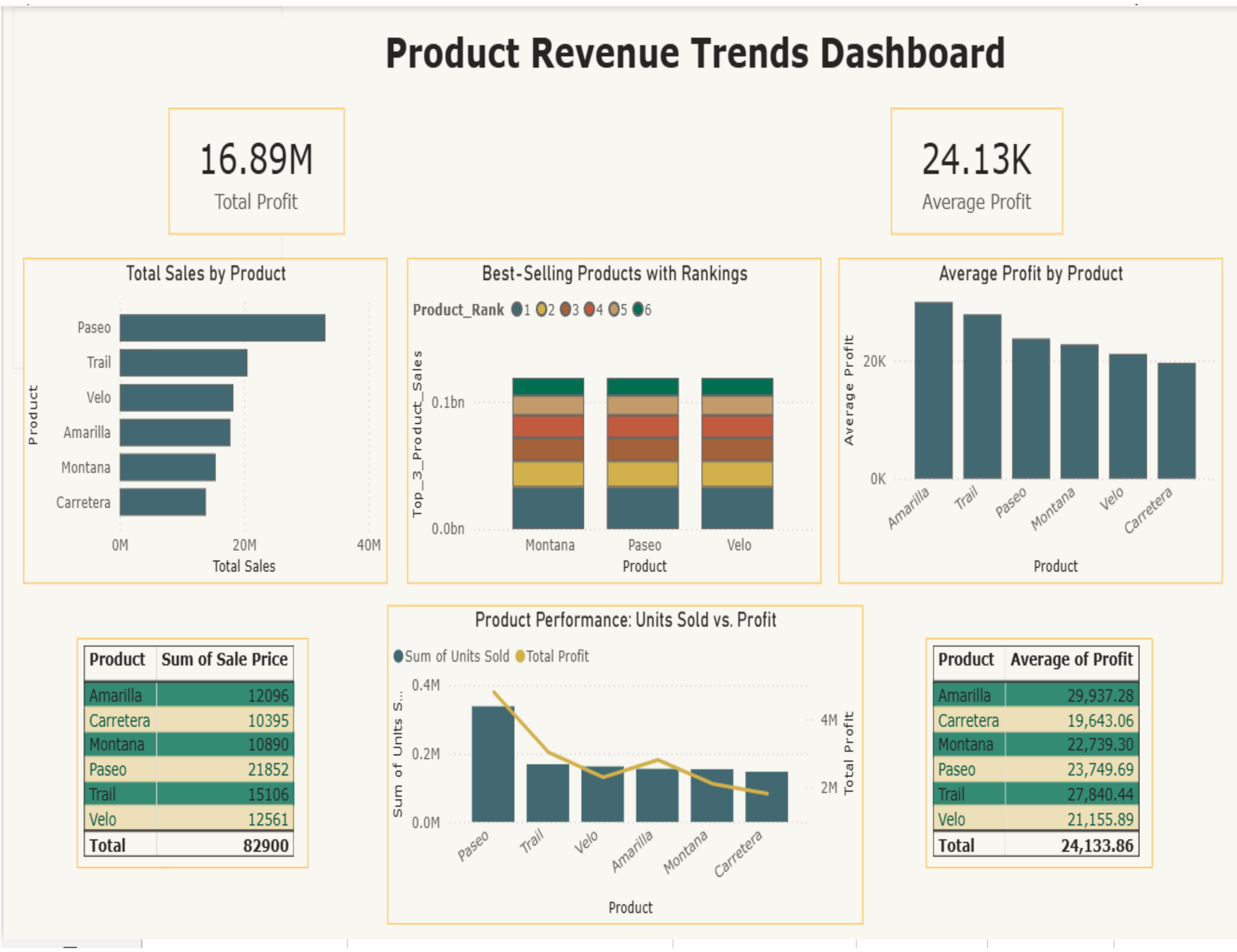
- Government, Small Business, USA, and Canada will remain key growth drivers.
- Paseo and Trail are expected to stay top performers in sales and profit.
- Channel Partners, Midmarket, Mexico, and Carretera may continue underperforming unless costs, pricing, or reach improve.
- Enterprise and weaker regions hold growth potential with targeted strategies.
- Sustaining investment in strong categories and monitoring costs will maximize long-term profitability

➢ Revenue Trends and Sales Insights



Business Data Analysis and Reporting

➢ Product Revenue Trends Dashboard



➢ Conclusion

The dashboard provides a clear view of overall revenue trends and product-level performance, enabling quick identification of top-performing categories and areas needing improvement. By combining sales and profit insights with product rankings, the analysis supports better decision-making on pricing, marketing, and resource allocation.

In short, the Power BI dashboard transforms raw sales data into actionable intelligence, helping stakeholders track growth, diagnose challenges, forecast future outcomes, and prescribe strategies to maximize profitability